

TEMPERATURE CONTROLLED FAN WITH ARDUINO

A Mini Project Report submitted

to fulfill the project-based learning Methodology

in

II-Year II-Semester

By

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

**RAMACHANDRA COLLEGE OF ENGINEERING
(AUTONOMOUS)**

(APPROVED BY AICTE - ACCREDITED BY NBA)

ELURU – 534007

(APRIL-2026)

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CERTIFICATE



This is to certify that the project report entitled "**Temperature controlled fan with Ardino**" that Being submitted by **P. Purna Samhitha (24ME1A04J7)** to fulfill the Mini project in Electronics and Communication Engineering to the Ramachandra College of Engineering, Eluru affiliated to JNTUK, Kakinada is a record of bonafide work carried out during the AY 2025-26.

Project Guide

Mrs.Ch.Lakshmi

Head of the department

Dr. B. Raghavaiah.

Evaluator

ACKNOWLEDGEMENT

I would like to take this opportunity to express my deepest appreciation to the faculty coordinators for their valuable contributions and assistance with this project based learning

I would like to thank **Mrs.Ch.Lakshmi** for his guidance and support, especially for the valuable ideas and knowledge provided throughout the project, valuable comments and suggestions have been very useful in solving problems encountered during the project.

I wish to express our sincere thanks to **Dr. B. Raghavaiah**, Professor and Head, Department of Electronics and Communication Engineering.

P. Purna Samhitha

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DECLARATION

I hereby declare that the work present in this project entitled "**TEMPERATURE CONTROLLED FAN WITH ARDUINO**" submitted towards the fulfilment of the requirements for the mini project (project-based learning Methodology) in Electronics and Communication in Ramachandra College of Engineering, Eluru is an authentic record of my work carried out for mini project, ECE Department, Ramachandra College of Engineering. The matter embodied in this report has not been submitted by us for the award of any other degree.

P. Purna Samhitha

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ABSTRACT

This mini project presents the design and implementation of an automatic temperature-controlled fan system using an Arduino microcontroller. The primary objective is to regulate fan operation based on surrounding temperature conditions without human intervention. A temperature sensor (such as LM35 or DHT11) is used to continuously monitor ambient temperature and send analog/digital signals to the Arduino. The microcontroller processes this data and compares it with predefined threshold values.

When the temperature exceeds the set limit, the Arduino activates the fan using a transistor or relay module. In advanced implementations, Pulse Width Modulation (PWM) is used to vary the fan speed proportionally with temperature changes, ensuring efficient cooling and energy conservation. At lower temperatures, the fan remains off or runs at reduced speed, minimizing power consumption.

This system is cost-effective, energy-efficient, and suitable for applications in homes, offices, and industrial environments where temperature regulation is essential. The project demonstrates the practical use of embedded systems, sensor interfacing, and automation techniques. It can be further enhanced by integrating an LCD display for real-time temperature monitoring or IoT modules for remote control and data logging.

Overall, the project provides a simple yet effective solution for automatic temperature-based cooling, highlighting the importance of smart systems in modern electronic applications.

INTRODUCTION

Automation plays a crucial role in modern electronic systems by improving efficiency, accuracy, and energy management. One common application of automation is temperature control, which is essential in homes, offices, industries, and electronic devices to maintain safe and comfortable conditions. Manual control of fans is often inefficient, as it depends on human intervention and may lead to unnecessary power consumption.

This mini project focuses on designing an Arduino-based temperature controlled fan that operates automatically according to the surrounding temperature. The system uses a temperature sensor such as LM35 or DHT11 to continuously measure ambient temperature. The sensed data is sent to the Arduino microcontroller, which processes the input and decides whether to turn the fan ON, OFF, or adjust its speed.

The Arduino acts as the central control unit, making decisions based on predefined temperature thresholds. When the temperature rises above a certain level, the fan is activated to provide cooling. In more advanced implementations, the fan speed can be varied using Pulse Width Modulation (PWM), allowing efficient energy utilization and better temperature regulation.

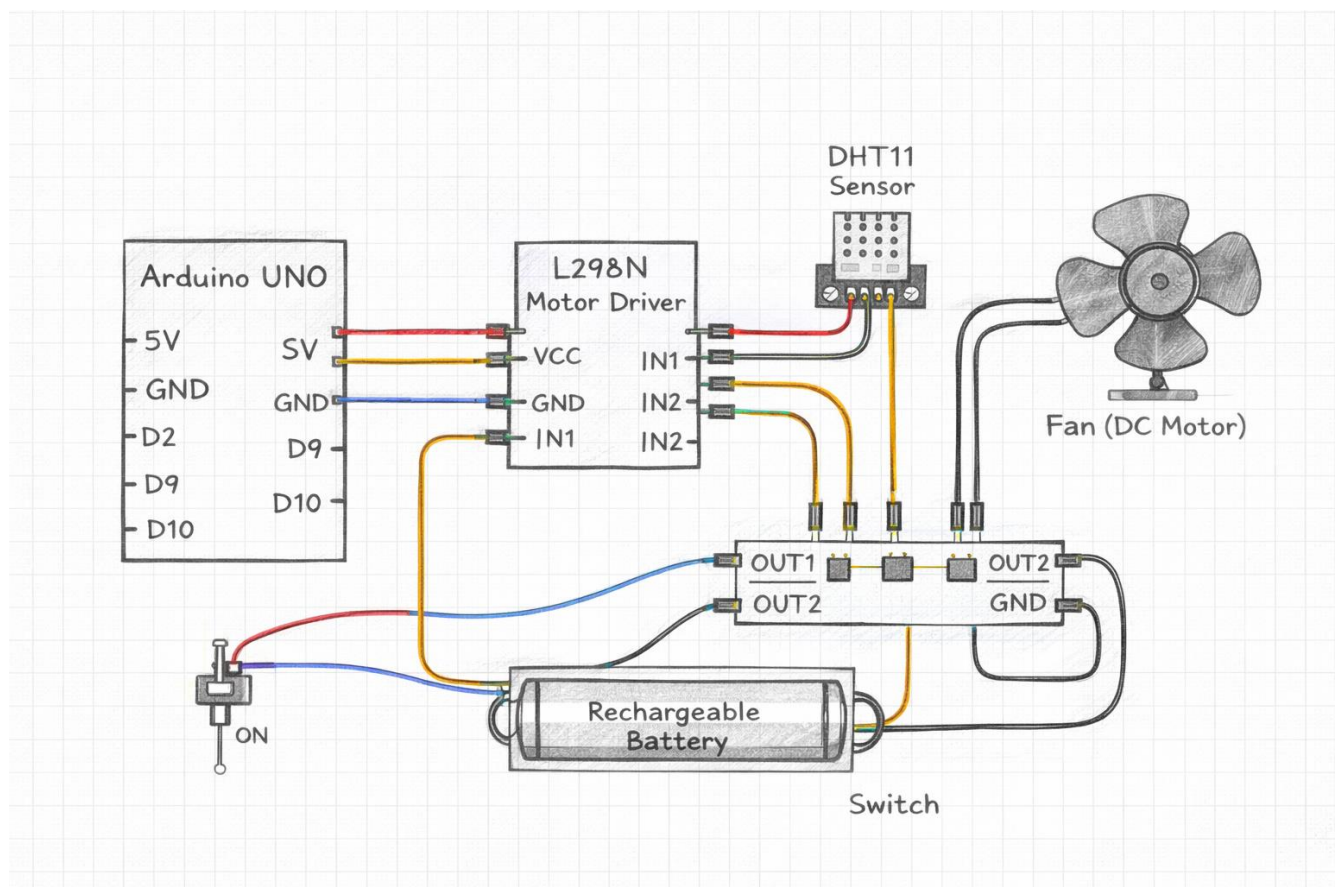
This project demonstrates the integration of sensors, microcontrollers, and actuators to create an intelligent system. It also helps in understanding key concepts such as analog-to-digital conversion, real-time monitoring, and embedded system design. Due to its simplicity, low cost, and effectiveness, this system can be widely used in practical applications such as automatic cooling systems, electronic device protection, and smart home solutions.

Overall, the Arduino-based temperature controlled fan provides a reliable and energy-efficient solution for automatic temperature management, showcasing the importance of embedded systems in everyday life.

Components:

- Arduino uno
- L298N Motor driver
- DHT11 Temperature sensor
- Fan (DC Motor)
- Switch
- Jumper wires
- Rechargeable

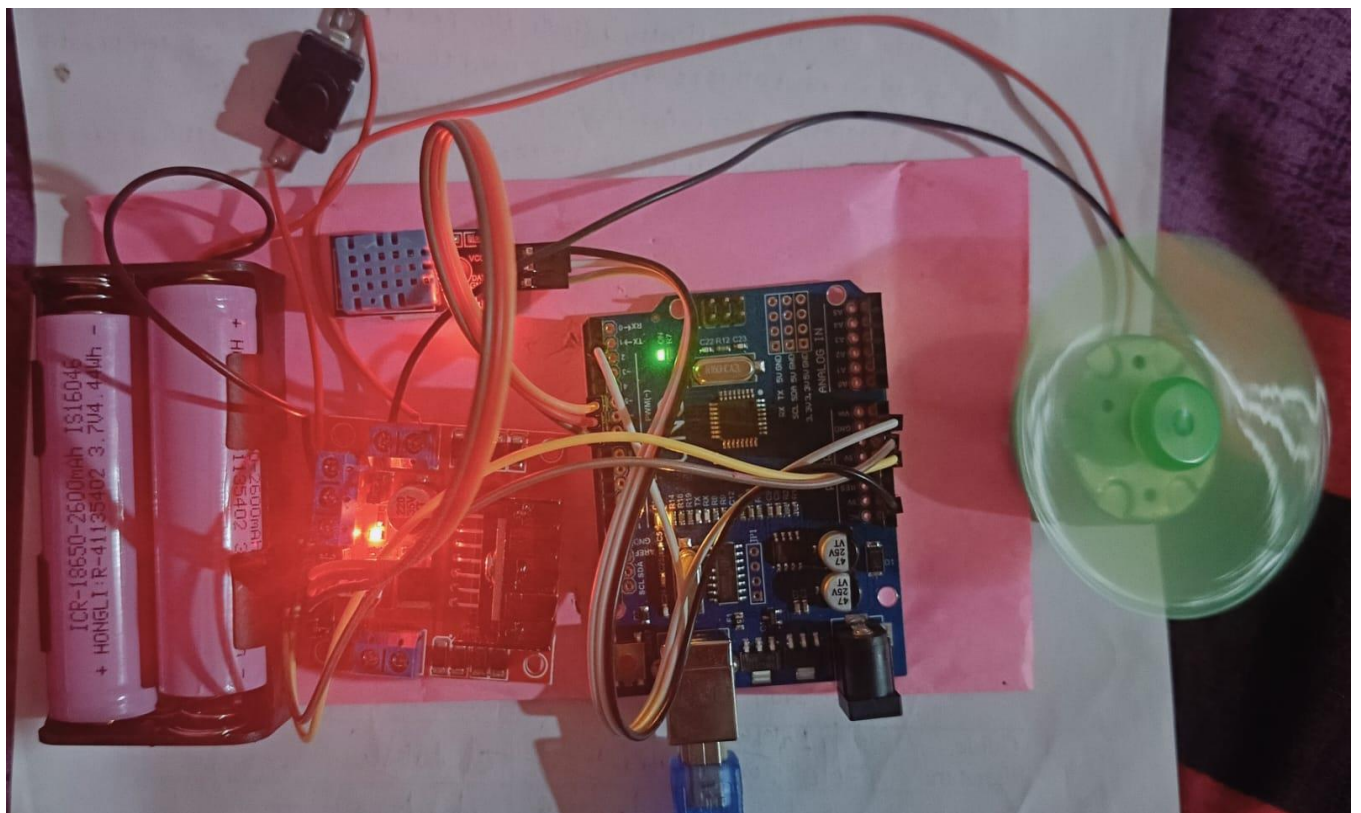
Circuit diagram:



Circuit description

Overview:

The Arduino-based temperature controlled fan system is an automatic cooling solution that operates based on real-time temperature conditions. The system consists of three main sections: input (temperature sensing), processing (Arduino), and output (fan control). The temperature sensor continuously monitors the surrounding temperature and sends corresponding signals to the Arduino microcontroller. Based on predefined conditions, the Arduino controls the fan, ensuring efficient cooling and reduced power consumption.



Working and output:

Working:

The temperature sensor (LM35/DHT11) continuously measures the ambient temperature and sends the signal to the Arduino. The Arduino reads this data and compares it with a preset temperature value. If the temperature exceeds the threshold, the Arduino sends a signal to a transistor or relay, which turns ON the fan. If the temperature is below the set limit, the fan remains OFF. In advanced cases, PWM is used to control fan speed based on temperature.

Output:

1. Low Temperature Condition

- Temperature below threshold (e.g., $< 30^{\circ}\text{C}$)
- Fan remains OFF
- Power consumption is minimized

2. Moderate Temperature Condition

- Temperature within mid-range (e.g., 30°C – 40°C)
- Fan turns ON at low or medium speed
- Maintains comfortable temperature

3. High Temperature Condition

- Temperature above threshold (e.g., $> 40^{\circ}\text{C}$)
- Fan runs at full speed
- Provides maximum cooling

4. Dynamic Output (PWM Control)

- Fan speed changes continuously with temperature
- Smooth and efficient cooling
- Avoids sudden ON/OFF switching

5. Display Output (if LCD used)

- Shows temperature (e.g., “Temp: 32°C ”)
- Displays fan status (e.g., “Fan: ON”)

Code:

```

#include <DHT.h>

// DHT Sensor Setup
#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

// Motor Driver Pins (L298N)
#define ENA 5
#define IN1 6
#define IN2 7

// Temperature Range
#define MIN_TEMP 25
#define MAX_TEMP 40

void setup() {
  Serial.begin(9600);
  dht.begin();

  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  // Fan Direction
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
}

void loop() {
  float temp = dht.readTemperature();

  // Check sensor error
  if (isnan(temp)) {
    Serial.println("DHT Error!");
    delay(2000);
    return;
  }

  Serial.print("Temperature: ");
  Serial.print(temp);
  Serial.println(" °C");

  int fanSpeed = 0;

  if (temp <= MIN_TEMP) {
    fanSpeed = 0; // OFF
  }
  else if (temp >= MAX_TEMP) {
    fanSpeed = 255; // FULL SPEED
  }
  else {
    fanSpeed = map((int)temp, MIN_TEMP, MAX_TEMP, 0, 255);
  }

  analogWrite(ENA, fanSpeed);

  Serial.print("Fan Speed (PWM): ");
  Serial.println(fanSpeed);

  delay(2000);
}

```

Applications:

1. Home Automation Systems

Used in smart homes to automatically control fans based on room temperature, improving comfort and reducing unnecessary power consumption.

2. Electronic Device Cooling

Applied in computers, laptops, power supplies, and embedded systems to prevent overheating and ensure proper functioning of components.

3. Industrial Temperature Control

Used in industries to maintain safe operating temperatures of machines, motors, and equipment, preventing damage and improving efficiency.

4. Server Rooms and Data Centers

Helps in automatic cooling of servers and networking equipment, ensuring system stability and preventing hardware failures due to overheating.

5. Smart Agriculture Systems

Used in greenhouses to regulate temperature, creating a suitable environment for plant growth and improving crop yield.

6. Refrigeration and Ventilation Systems

Integrated into ventilation and cooling systems to maintain desired temperature levels automatically.

7. Automotive Applications

Used in vehicles for engine cooling systems and cabin temperature control.

8. Laboratories and Medical Equipment

Maintains controlled temperature conditions for sensitive instruments and storage systems.

9. Offices and Commercial Buildings

Provides automated cooling in workspaces, reducing manual control and improving energy efficiency.

10. IoT-Based Smart Systems

Can be integrated with IoT platforms for remote monitoring and control of temperature and fan operation through mobile or web applications.

Future Scope:

- **IoT Integration:** Connect the system to cloud platforms for remote monitoring and control using mobile apps.
- **AI-Based Automation:** Implement machine learning to predict temperature patterns and adjust fan speed intelligently.
- **Energy Optimization:** Integrate solar power or smart energy-saving modes to reduce electricity consumption.
- **Multi-Sensor Expansion:** Add humidity, air quality, or motion sensors for a more advanced environmental control system.
- **Smart Home Integration:** Interface with voice assistants (like Alexa/Google Assistant) for automated control.

Conclusion:

The Arduino-based temperature controlled fan is a simple and effective system for automatic temperature regulation. It uses a temperature sensor and Arduino to control the fan based on environmental conditions, reducing the need for manual operation. The system is energy-efficient, cost-effective, and reliable. It demonstrates basic concepts of embedded systems and can be widely used in home, industrial, and electronic cooling applications.